

		$(1.0)(9.81)(1.0) = KE + (0.5)(10.0)$ $KE = 4.81 \text{ N}$
8	B	Let the extension of the spring when it is at equilibrium be x Loss in GPE = mgx Gain in EPE = $\frac{1}{2} kx^2$ Since the spring is in equilibrium, $F = kx = mg$ Hence gain in EPE = $\frac{1}{2} (kx)x = \frac{1}{2} mgx = \frac{1}{2} \text{ GPE}$
9	D	Maximum tension experienced at bottom of the swing. Centripetal force = tension – weight